MATHS MANTHAN
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CBSE CLASS 10 | MATHEMATICS

POLYNOMIALS

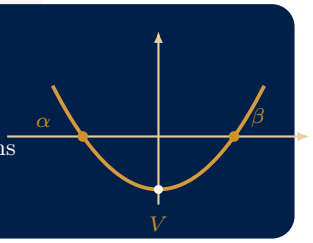
Master Concept, Formula & Visual Problem-Solving Handbook

Academic Year 2026–27

Identify the degree. Visualize the curve.
Connect the roots to the coefficients.

Geometric Zeroes | Parabola Discriminant Cases | Vieta's Relations | Symmetric Functions

A student-friendly Maths Manthan revision resource



What this handbook includes

- Complete geometry of zeroes: Line and Parabola analysis.
- The 3 Discriminant cases ($D > 0, D = 0, D < 0$) in 6 orientations.
- Vieta's formulas: $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.
- The 8 master symmetric identity expansions for board PYQs.
- Parameter conditions: Reciprocal, opposite, and ratio roots.

How to use this handbook

Read the degree of the polynomial first. On graphs, count only the **intersections with the x-axis**. When forming quadratic polynomials, always include the non-zero real multiplier k ($p(x) = k[x^2 - Sx + P]$) to clear denominators. Evaluate symmetric expressions using S and P directly without solving for roots.

Official CBSE Curriculum Scope (Academic Year 2026–27)

Strict Syllabus Integrity & Deleted Content: Under the rationalized CBSE 2026–27 curriculum, the **Division Algorithm for Polynomials** ($p(x) = g(x)q(x) + r(x)$) and polynomial long division (former NCERT Exercise 2.3 and Exercise 2.4) have been **permanently excised** from Chapter 2. This handbook strictly covers the complete 100% prescribed syllabus: Zeroes of polynomials, Geometrical meaning, and the Relationship between zeroes and coefficients of quadratic polynomials. Source: CBSE Secondary Curriculum (2026–27), Unit II: Algebra.

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Prepared for learning and revision – not an official CBSE publication.

CHAPTER 02

SUPPORTING SKILL

ALGEBRAIC FOUNDATIONS: POLYNOMIALS

Definition, Classification by Degree and Terms & Diagnostic Identification

Module 1.1 | What is a Polynomial?

An algebraic expression $p(x)$ in one variable x is called a **polynomial** if all exponents of the variable x are **non-negative integers** (whole numbers $\mathbb{W} = \{0, 1, 2, 3, \dots\}$):

$$\boxed{p(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0}$$

where $a_n, a_{n-1}, \dots, a_0 \in \mathbb{R}$ are real constants (coefficients) and $a_n \neq 0$. The highest exponent n is the **Degree** of the polynomial.

Module 1.2 | The "Is It a Polynomial?" Diagnostic Test

Expression	Polynomial?	Mathematical Reason
$3x^2 - 5x + 7$	**YES**	All exponents of x are whole numbers $(2, 1, 0)$. Coefficients are real.
$\sqrt{3}x^2 + \frac{1}{2}x$	**YES**	Coefficients can be irrational or fractional. Powers of x are integers $(2, 1)$.
$2\sqrt{x} + 5$	**NO**	Exponent of x is fractional: $x^{1/2} \notin \mathbb{W}$.
$x^2 + \frac{3}{x} - 4$	**NO**	Variable in denominator implies negative exponent: $x^{-1} \notin \mathbb{W}$.
$\frac{x^2-4}{x-2}$	**NO**	Rational expression with a domain restriction ($x \neq 2$). Not globally defined.

Module 1.3 | Classification by Degree & Term Count



Module 1.4 | Value of a Polynomial $p(k)$ vs. Zero of a Polynomial

- **Value at $x = k$:** Obtained by substituting $x = k$ into $p(x)$, denoted as $p(k)$.
- **Zero of a Polynomial:** A real number k is called a **zero** of $p(x)$ if and only if $\mathbf{p(k) = 0}$.
- **Fundamental Principle:** A polynomial of degree n has **at most n real zeroes**.

CHAPTER 02

CORE SYLLABUS

GEOMETRIC MEANING OF ZEROES

The Fundamental Coordinate Principle, Linear Lines & Parabola Anatomy

Module 2.1 | The Fundamental Coordinate-Geometry Principle

The real zeroes of a polynomial $p(x)$ are precisely the **x-coordinates of the points where the graph of $y = p(x)$ intersects (or touches) the x-axis**.

$$\boxed{p(k) = 0 \iff (k, 0) \in \text{Graph of } y = p(x)}$$

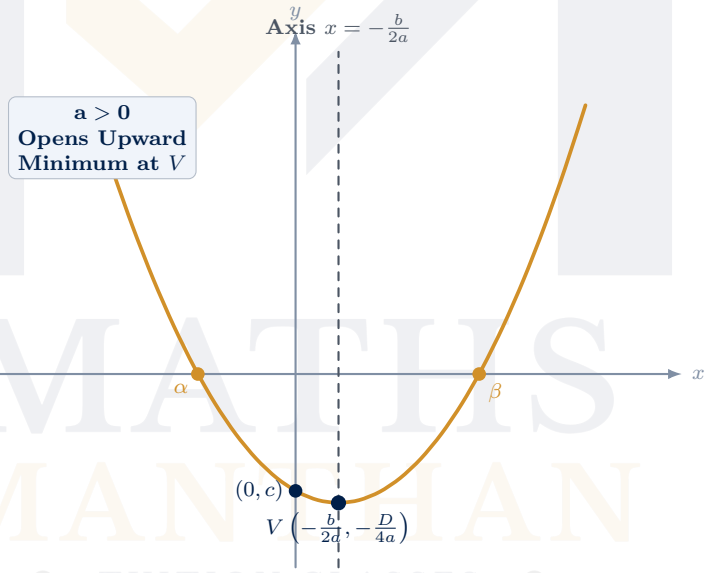
Module 2.2 | Graph of a Linear Polynomial ($y = ax + b,\ a \neq 0$)

The graph of a linear polynomial is a straight line. Since a non-horizontal line intersects the x-axis at exactly one point, a linear polynomial has **exactly one zero**: $\mathbf{x = -\frac{b}{a}}$.

Module 2.3 | Graph of a Quadratic Polynomial ($y = ax^2 + bx + c$): The Parabola

The graph of any quadratic polynomial is a continuous, U-shaped symmetrical curve called a **Parabola**.

- **Orientation:** Opens **upward** ($\bigcup$) if $a > 0$; opens **downward** ($\bigcap$) if $a < 0$.
- **Axis of Symmetry:** The vertical line passing through the midpoint of the zeroes: $\mathbf{x = -\frac{b}{2a}}$.
- **Vertex (V):** The turning point (minimum when $a > 0$; maximum when $a < 0$): $\mathbf{V\left(-\frac{b}{2a}, -\frac{D}{4a}\right)}$.
- **y-Intercept:** Setting $x = 0$ gives $y = c$, so the graph always crosses the y-axis at $\mathbf{(0, c)}$.



Geometric Core Rule: How Many Zeroes Does a Curve Have?

To determine the number of zeroes of any polynomial from its graph, count the number of times the curve **cuts or touches the x-axis**. Intersections with the y-axis are irrelevant to zeroes!

CHAPTER 02

CORE SYLLABUS

THE THREE DISCRIMINANT CASES

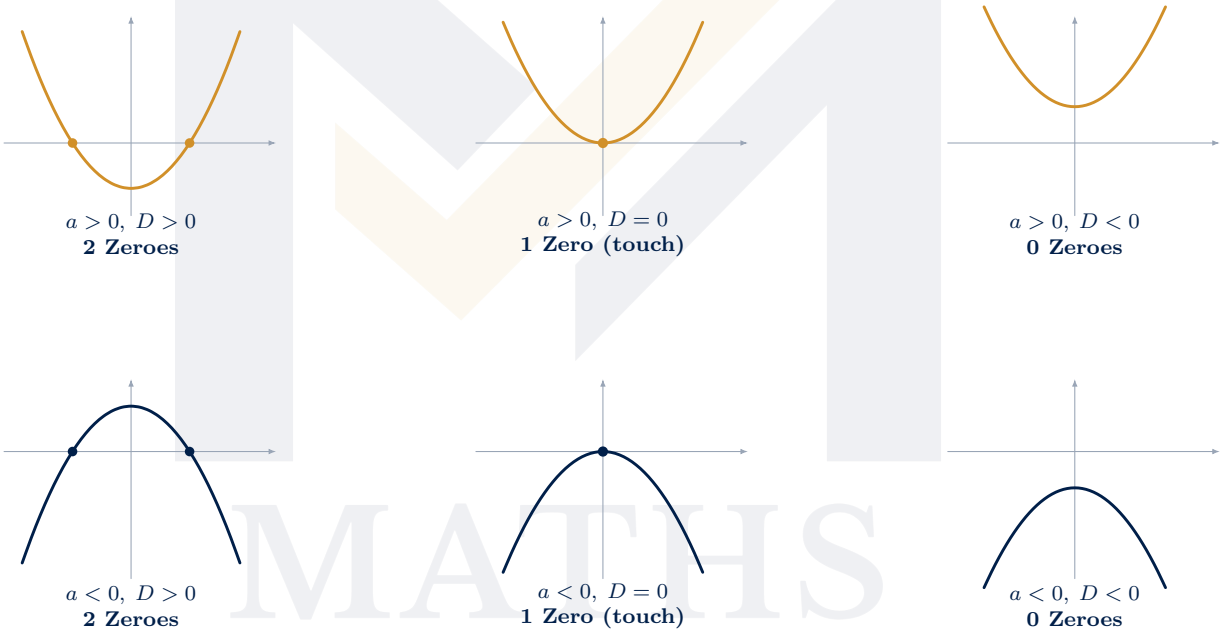
The 6-Parabola Matrix, Intersections vs. Contacts & Graph Reading Protocols

Module 3.1 | The Discriminant ($D = b^2 - 4ac$) as the Geometric Arbiter

The sign of the discriminant $D = b^2 - 4ac$ uniquely determines how many times the parabola meets the x-axis:

Case	Discriminant	Common Points	Real Zeroes	Geometric Behavior of Parabola
I	**$D > 0$**	2	2 distinct	Parabola cuts cleanly through x-axis at $(\alpha, 0)$ and $(\beta, 0)$.
II	**$D = 0$**	1 (tangent)	1 repeated	Parabola touches the x-axis at its vertex $V\left(-\frac{b}{2a}, 0\right)$.
III	**$D < 0$**	0	0 real	Parabola does not meet x-axis at all (lies entirely above or below).

Module 3.2 | The Six-Parabola Matrix



Module 3.3 | NCERT Exercise 2.1 Graph-Reading Protocol

Board Exam 1-Mark MCQ Protocol: Reading Curves

When a question asks *"Find the number of zeroes of $y = p(x)$ from the given figure"*:

1. Focus **exclusively on the horizontal x-axis**. Ignore where the curve intersects the y-axis.
2. Each distinct crossing counts as 1 zero. Each point of tangency (touching without crossing) counts as 1 repeated real zero.
3. If the curve does not meet the x-axis, write: **"Number of zeroes = 0"**.

CHAPTER 02

CORE SYLLABUS

ZEROES & COEFFICIENTS: VIETA'S RELATIONS

Factor Theorem Derivation, Verification Protocols & Parabola Symmetry

Module 4.1 | Factor Theorem Derivation of Vieta's Relations

Let $p(x) = ax^2 + bx + c$ $(a \neq 0)$ have two real zeroes α and β. By the **Factor Theorem**, $(x - \alpha)$ and $(x - \beta)$ are factors of $p(x)$:

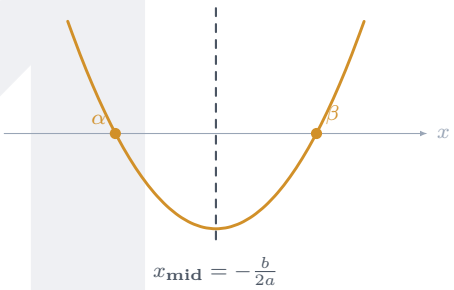
$$ax^2 + bx + c = k(x - \alpha)(x - \beta) = k\left[x^2 - (\alpha + \beta)x + \alpha\beta\right] = kx^2 - k(\alpha + \beta)x + k\alpha\beta$$

Equating coefficients of like powers of x:

$$a = k, \qquad b = -a(\alpha + \beta) \implies \alpha + \beta = -\frac{\mathbf{b}}{\mathbf{a}}, \qquad c = a\alpha\beta \implies \alpha\beta = \frac{\mathbf{c}}{\mathbf{a}}$$

$$\boxed{\mathbf{S} = \alpha + \beta = -\frac{\mathbf{b}}{\mathbf{a}} = -\frac{\text{Coeff. of } \mathbf{x}}{\text{Coeff. of } \mathbf{x^2}}} \qquad \boxed{\mathbf{P} = \alpha\beta = \frac{\mathbf{c}}{\mathbf{a}} = \frac{\text{Constant term}}{\text{Coeff. of } \mathbf{x^2}}}$$

Module 4.2 | Geometric Interpretation: The Symmetry of Roots



Notice that the axis of symmetry of the parabola $y = ax^2 + bx + c$ is located at the exact arithmetic midpoint of the two zeroes:

$$x_{\text{mid}} = \frac{\alpha + \beta}{2} = \frac{-\frac{b}{a}}{2} = -\frac{\mathbf{b}}{\mathbf{2a}}$$

Thus, Vieta's sum of zeroes is directly encoded as the horizontal location of the parabola's vertex!

Module 4.3 | Worked Archetype 1: Finding Zeroes & Verifying Relations

Worked Archetype 1: Complete CBSE Board 3-Marker Protocol

Problem: Find zeroes of $p(x) = 6x^2 - 3 - 7x$ and verify relations with coefficients.
Solution:

1. **Standard Form:** $p(x) = 6x^2 - 7x - 3$. Here $a = 6, b = -7, c = -3$.
2. **Factorization:** $6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (2x - 3)(3x + 1)$.
3. Zeroes: $2x - 3 = 0 \implies \alpha = \mathbf{\frac{3}{2}}$ and $3x + 1 = 0 \implies \beta = \mathbf{-\frac{1}{3}}$.
4. **Verification:** $\alpha + \beta = \frac{3}{2} - \frac{1}{3} = \frac{7}{6}$ and $-\frac{b}{a} = -\frac{-7}{6} = \frac{7}{6}$ [Verified].
   $\alpha\beta = \left(\frac{3}{2}\right)\left(-\frac{1}{3}\right) = -\frac{1}{2}$ and $\frac{c}{a} = \frac{-3}{6} = -\frac{1}{2}$ [Verified].

CHAPTER 02

CORE SYLLABUS

FORMATION OF QUADRATIC POLYNOMIALS

Reconstructing Polynomials from Zeroes & The Vital Role of Constant k

Module 5.1 | The General Reconstruction Formula

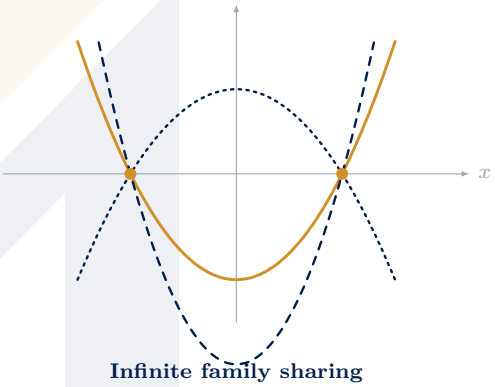
If α and β are the zeroes of a quadratic polynomial, the polynomial can be reconstructed uniquely up to a non-zero scaling factor k:

$$\boxed{p(x) = k\left[x^2 - (\alpha+\beta)x + \alpha\beta\right] = \mathbf{k\left[x^2 - Sx + P\right]}}$$

where $S = \alpha + \beta$ is the **Sum**, $P = \alpha\beta$ is the **Product**, and $k \in \mathbb{R} \setminus \{0\}$.

Module 5.2 | Why the Constant k is Mandatory: The Family of Polynomials

A pair of zeroes does not specify a single unique polynomial, but an **infinite family of scaled polynomials** having identical x-intercepts:



Infinite family sharing roots α and β

Example with Fractional Coefficients:
Suppose $S = -\frac{1}{4}$ and $P = \frac{1}{4}$.

$$p(x) = k\left[x^2 - \left(-\frac{1}{4}\right)x + \frac{1}{4}\right] = k\left[x^2 + \frac{1}{4}x + \frac{1}{4}\right]$$

- Setting $k = 4$ clears the fractions to give integer coefficients:

$$\mathbf{p(x) = 4x^2 + x + 1}$$

- Both $x^2 + \frac{1}{4}x + \frac{1}{4}$ and $4x^2 + x + 1$ share the exact same zeroes!

Module 5.3 | Worked Archetypes: Forming Quadratic Polynomials

Worked Archetype 2: Given Zeroes vs. Given Sum and Product

Case A (Direct Sum and Product): Find a quadratic polynomial with sum $S = \sqrt{2}$ and product $P = \frac{1}{3}$.
Solution: $p(x) = k\left[x^2 - Sx + P\right] = k\left[x^2 - \sqrt{2}x + \frac{1}{3}\right]$. Setting $k = 3$: $\mathbf{p(x) = 3x^2 - 3\sqrt{2}x + 1}$.

Case B (Individual Zeroes Given): Find a quadratic polynomial whose zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.
Solution:

- $S = (2+\sqrt{3}) + (2-\sqrt{3}) = 4$.
- $P = (2+\sqrt{3})(2-\sqrt{3}) = 2^2 - (\sqrt{3})^2 = 4 - 3 = 1$.
- $p(x) = k[x^2 - 4x + 1]$. For $k = 1$: $\mathbf{p(x) = x^2 - 4x + 1}$.

CHAPTER 02

CORE SYLLABUS

SYMMETRIC FUNCTIONS OF ZEROES

The 8 Core Algebraic Transformations & Identity Derivations in S and P

Module 6.1 | The Principle of Symmetric Functions

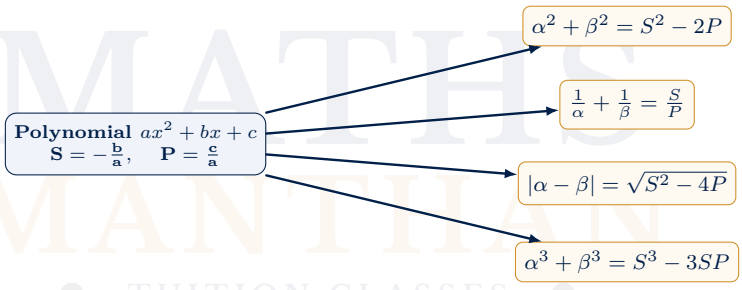
An algebraic expression in α and β is called **symmetric** if swapping α and β leaves the expression unchanged ($f(\alpha, \beta) = f(\beta, \alpha)$).

The Golden Rule of Board Evaluation: Never calculate α and β individually! Express symmetric functions strictly in terms of $\mathbf{S} = \alpha + \beta = -\frac{\mathbf{b}}{\mathbf{a}}$ and $\mathbf{P} = \alpha\beta = \frac{\mathbf{c}}{\mathbf{a}}$.

Module 6.2 | The 8 Essential Algebraic Identities

#	Expression	Identity in S and P	Derivation / Expansion Blueprint
1	$\alpha^2 + \beta^2$	$\mathbf{S^2 - 2P}$	From $(\alpha+\beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$.
2	$\frac{1}{\alpha} + \frac{1}{\beta}$	$\frac{\mathbf{S}}{\mathbf{P}}$	Take common denominator: $\frac{\beta+\alpha}{\alpha\beta} = \frac{S}{P}$.
3	$\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$	$\frac{\mathbf{S^2-2P}}{\mathbf{P}}$	Combine: $\frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{S^2-2P}{P}$.
4	$(\alpha - \beta)^2$	$\mathbf{S^2 - 4P}$	$(\alpha+\beta)^2 - 4\alpha\beta = S^2 - 4P \implies \lvert\alpha-\beta\rvert = \sqrt{S^2-4P}$.
5	$\alpha^3 + \beta^3$	$\mathbf{S^3 - 3SP}$	From $(\alpha+\beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha+\beta)$.
6	$\alpha^2\beta + \alpha\beta^2$	$\mathbf{PS}$	Factor out product: $\alpha\beta(\alpha+\beta) = P \cdot S$.
7	$\frac{1}{\alpha^2} + \frac{1}{\beta^2}$	$\frac{\mathbf{S^2-2P}}{\mathbf{P^2}}$	$\frac{\alpha^2+\beta^2}{(\alpha\beta)^2} = \frac{S^2-2P}{P^2}$.
8	$\alpha^4 + \beta^4$	$\mathbf{(S^2 - 2P)^2 - 2P^2}$	Treat as $(\alpha^2+\beta^2)^2 - 2(\alpha\beta)^2$.

Module 6.3 | The Symmetric Function Architecture



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CHAPTER 02

WORKED MODEL

WORKED BOARD-EXAM MASTER MODELS

Expression Evaluations, Root Transformations & Finding Unknown Parameters

Module 7.1 | Master Model 1: Non-Factorable Expression Evaluation

Worked Archetype 3: Evaluating Complex Symmetric Expressions

Problem: If α and β are the zeroes of $p(x) = 2x^2 - 5x + 7$, find the value of: (i) $\frac{1}{\alpha} + \frac{1}{\beta}$ (ii) $\alpha^3 + \beta^3$.
Solution: Here $a = 2, b = -5, c = 7$. $S = \alpha + \beta = -\frac{-5}{2} = \frac{5}{2}, \quad P = \alpha\beta = \frac{7}{2}$.

1. $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{\frac{5}{2}}{\frac{7}{2}} = \mathbf{\frac{5}{7}}$.
2. $\alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta) = \left(\frac{5}{2}\right)^3 - 3\left(\frac{7}{2}\right)\left(\frac{5}{2}\right) = \frac{125}{8} - \frac{105}{4} = \frac{125-210}{8} = -\mathbf{\frac{85}{8}}$.

Module 7.2 | Master Model 2: Forming Polynomials with Transformed Zeroes

Worked Archetype 4: Polynomial with Reciprocal Zeroes

Problem: If α, β are zeroes of $p(x) = x^2 - 2x - 15$, form a quadratic polynomial whose zeroes are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.
Solution: For $p(x)$, $\alpha + \beta = 2$ and $\alpha\beta = -15$. Let S' and P' be the sum and product of the new zeroes:

- $S' = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{2}{-15} = -\frac{2}{15}$.
- $P' = \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{\alpha\beta} = -\frac{1}{15}$.
- Required polynomial: $q(x) = k[x^2 - S'x + P'] = k\left[x^2 - \left(-\frac{2}{15}\right)x + \left(-\frac{1}{15}\right)\right] = k\left[x^2 + \frac{2}{15}x - \frac{1}{15}\right]$.
- Setting $k = 15$ gives: $\mathbf{q(x) = 15x^2 + 2x - 1}$.

Module 7.3 | Master Model 3: Finding Unknown Parameter k

Worked Archetype 5: Parameter with Zero-Difference Condition

Problem: If the zeroes of $p(x) = x^2 - 8x + k$ satisfy $(\alpha - \beta) = 2$, find the value of k.
Solution: Here $a = 1, b = -8, c = k \implies \alpha + \beta = 8$ and $\alpha\beta = k$. Using identity: $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$:

$$(2)^2 = (8)^2 - 4k \implies 4 = 64 - 4k \implies 4k = 60 \implies \mathbf{k = 15}.$$

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CHAPTER 02

WORKED MODEL

SPECIAL ZERO CONFIGURATIONS

Reciprocal Roots, Additive Inverses & Rapid Elimination Shortcuts

Module 8.1 | Reciprocal Zeroes Theorem ($\beta = \frac{1}{\alpha}$)

If one zero of $p(x) = ax^2 + bx + c$ is the **reciprocal** of the other:

$$\alpha \cdot \beta = 1 \implies \frac{c}{a} = 1 \implies \mathbf{c = a}$$

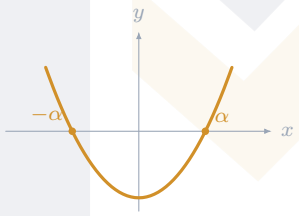
Board Shortcut: Whenever a question states "one zero is reciprocal of the other", directly equate Constant term = Coefficient of $\mathbf{x^2}$.

Module 8.2 | Equal and Opposite Zeroes Theorem ($\beta = -\alpha$)

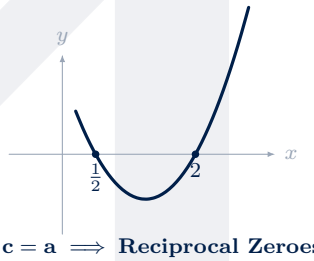
If one zero is the **negative (additive inverse)** of the other:

$$\alpha + \beta = 0 \implies -\frac{b}{a} = 0 \implies \mathbf{b = 0}$$

Geometric Implication: When $b = 0$, the axis of symmetry is $x = 0$ (the y-axis). The parabola is perfectly symmetric about the y-axis!



b = 0 $\implies$ Opposite Zeroes
Symmetric about y-axis



c = a $\implies$ Reciprocal Zeroes
Product of roots $\alpha\beta = 1$

Module 8.3 | Zeroes in a Given Ratio ($\alpha : \beta = m : n$)

If the zeroes of $ax^2 + bx + c$ are in the ratio $m : n$, then:

$$\boxed{\frac{(m+n)^2}{mn} = \frac{b^2}{ac}}$$

Proof: Let $\alpha = mk, \beta = nk$. Then $\alpha + \beta = (m+n)k = -\frac{b}{a}$ and $\alpha\beta = mnk^2 = \frac{c}{a}$. Dividing $(S)^2$ by P immediately yields the identity.

CHAPTER 02

EXTENSION PRACTICE

ENRICHMENT BRIDGE: CUBIC POLYNOMIALS

Higher-Degree Vieta Relations, Cubic Curve Geometry & Syllabus Demarcation

Module 9.1 | Vieta's Relations for Cubic Polynomials

A cubic polynomial has degree 3 and at most 3 real zeroes:

$$p(x) = ax^3 + bx^2 + cx + d \quad (a \neq 0)$$

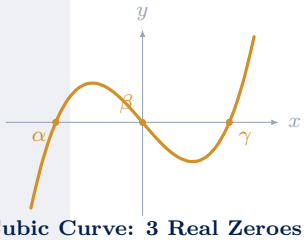
If α, β, γ are the zeroes of $p(x)$, then $p(x) = k(x-\alpha)(x-\beta)(x-\gamma)$. Expanding and equating coefficients yields:

Sum of Zeroes: $\alpha + \beta + \gamma = -\dfrac{\mathbf{b}}{\mathbf{a}} = -\dfrac{\text{coeff. of } x^2}{\text{coeff. of } x^3}$

Sum of Products taken two at a time: $\alpha\beta + \beta\gamma + \gamma\alpha = \dfrac{\mathbf{c}}{\mathbf{a}} = \dfrac{\text{coeff. of } x}{\text{coeff. of } x^3}$

Product of all three Zeroes: $\alpha\beta\gamma = -\dfrac{\mathbf{d}}{\mathbf{a}} = -\dfrac{\text{constant term}}{\text{coeff. of } x^3}$

Module 9.2 | Geometric Behavior of a Cubic Polynomial



Cubic Curve: 3 Real Zeroes

The graph of a cubic polynomial is an S-shaped curve that can intersect the x-axis at $1, 2$, or 3 points.

- A cubic polynomial always has **at least one real zero** (complex roots occur in conjugate pairs).
- Formation:

$$\mathbf{p(x) = k\left[x^3 - S_1x^2 + S_2x - S_3\right]}$$

where $S_1 = \sum \alpha,\ S_2 = \sum \alpha\beta,\ S_3 = \alpha\beta\gamma$.

CBSE 2026–27 Scope Demarcation Alert

The CBSE rationalized syllabus tests **quadratic polynomials exclusively** in the subjective board examinations. Cubic relations are provided here solely for competitive foundation (Olympiads/JEE Foundation) and MCQ conceptual completeness. Polynomial long division is strictly deleted.

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CHAPTER 02

REFERENCE MATRIX

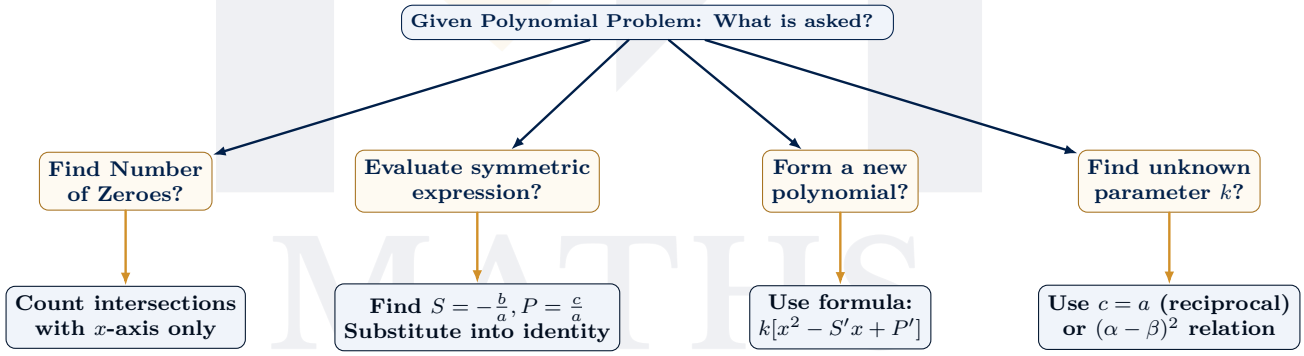
MASTER SUMMARY & DECISION ALGORITHM

Complete Formula Repository & 10-Second Problem Solver Flowchart

Module 10.1 | Comprehensive Polynomial Formula Repository

Concept / Formula	Mathematical Statement	Key Board Exam Application
Vieta's Sum	$\alpha + \beta = -\frac{b}{a}$	Direct evaluation; symmetric expressions; sign analysis.
Vieta's Product	$\alpha\beta = \frac{c}{a}$	Reciprocal roots check; product evaluations.
Polynomial Formation	$p(x) = k[x^2 - Sx + P]$	Creating quadratic polynomials from given zeroes or sum/product.
Sum of Squares	$\alpha^2 + \beta^2 = S^2 - 2P$	Most frequent 2-marker symmetric evaluation.
Difference of Zeroes	$(\alpha - \beta)^2 = S^2 - 4P$	Solving conditions where roots differ by a given constant.
Sum of Cubes	$\alpha^3 + \beta^3 = S^3 - 3SP$	Evaluating cubic root additions without solving for roots.
Reciprocal Zeroes	$c = a$	Instantly solving unknown coefficients when roots are reciprocals.
Opposite Zeroes	$b = 0$	Parabola symmetric across y-axis; roots sum to 0.

Module 10.2 | The 10-Second Problem-Solving Decision Flowchart



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CHAPTER 02

TRAP ALERT

EXAM TRAPS, HACKS & PRE-SUBMISSION AUDIT

Crucial Board Pitfalls, Instant Verification Checks & Final Protocol

Module 11.1 | Top 8 CBSE Board Exam Traps in Polynomials

1. **Forgetting the Negative Sign in Sum of Zeroes:** $\alpha + \beta = -\frac{b}{a}$, NOT $\frac{b}{a}$. If $b = -5$, then $-\frac{b}{a} = -\frac{-5}{a} = +\frac{5}{a}$.
2. **Omitting Multiplier k in Polynomial Formation:** Writing $p(x) = x^2 - Sx + P$ without the non-zero constant k loses marks when coefficients are fractional.
3. **Counting y-Axis Intersections on Graphs:** Zeroes are determined **solely by intersections with the x-axis**. Never count the point where the curve cuts the y-axis!
4. **Confusing "Number of Zeroes" with "Value of Zeroes":** If a parabola cuts the x-axis at $x = 3$ and $x = -2$, the **number** of zeroes is 2, while the **values** are 3 and -2.
5. **Non-Standard Term Ordering Trap:** In polynomials like $6x^2 - 3 - 7x$, students frequently write $b = -3$ and $c = -7$. Always rearrange into standard form $ax^2 + bx + c$ $(6x^2 - 7x - 3)$ first!
6. **Incorrect Expansion of $(\alpha - \beta)^2$:** Write $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$. Do NOT confuse this with $(\alpha + \beta)^2 - 2\alpha\beta$ (which is $\alpha^2 + \beta^2$).
7. **Attempting Long Division (Deleted Topic):** Do not use the deleted division algorithm $p(x) = g(x)q(x) + r(x)$ to solve questions. Use Vieta's formulas and symmetric identities.
8. **Zeroes of Polynomial vs. Roots of Equation:** A polynomial has **zeroes**; an equation has **roots**. Write $p(x) = 0$ when transitioning from polynomials to equations.

Module 11.2 | Algebraic Verification Hacks

- **Direct Substitution Check:** Substitute your calculated zero α back into $p(x)$. If $p(\alpha) \neq 0$, your root calculation is flawed.
- **Discriminant Reality Sanity Check:** Before looking for real zeroes, check $D = b^2 - 4ac$. If $D < 0$, no real zeroes exist.
- **Parity & Sign Check:** If $c/a > 0$, both zeroes must have the **same sign**. If $c/a < 0$, zeroes must have **opposite signs**.

Module 11.3 | The 5-Point Pre-Submission Board Exam Audit

Check	Verification Item	Board Evaluation Standard
☐ 1	Standard Form Arranged	Are terms written in strict descending powers of x before reading a, b, c?
☐ 2	Sign of $-\frac{b}{a}$ Verified	Did you properly apply the negative sign to Sum $= -\frac{b}{a}$?
☐ 3	Multiplier k Included	Did you write $p(x) = k[x^2 - Sx + P]$ with $k \neq 0$ stated?
☐ 4	x-Axis Intersections Counted	On graph questions, did you count only x-axis intersections and ignore y-axis?
☐ 5	Symmetric Identities Verified	Did you express symmetric expressions strictly in terms of S and P?

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